

TECHNICAL CASE STUDY

Care Gap Explorer

From healthcare data to actionable A1C monitoring gaps

116

patients in the diabetes cohort

25

open A1C monitoring gaps

365

day monitoring window

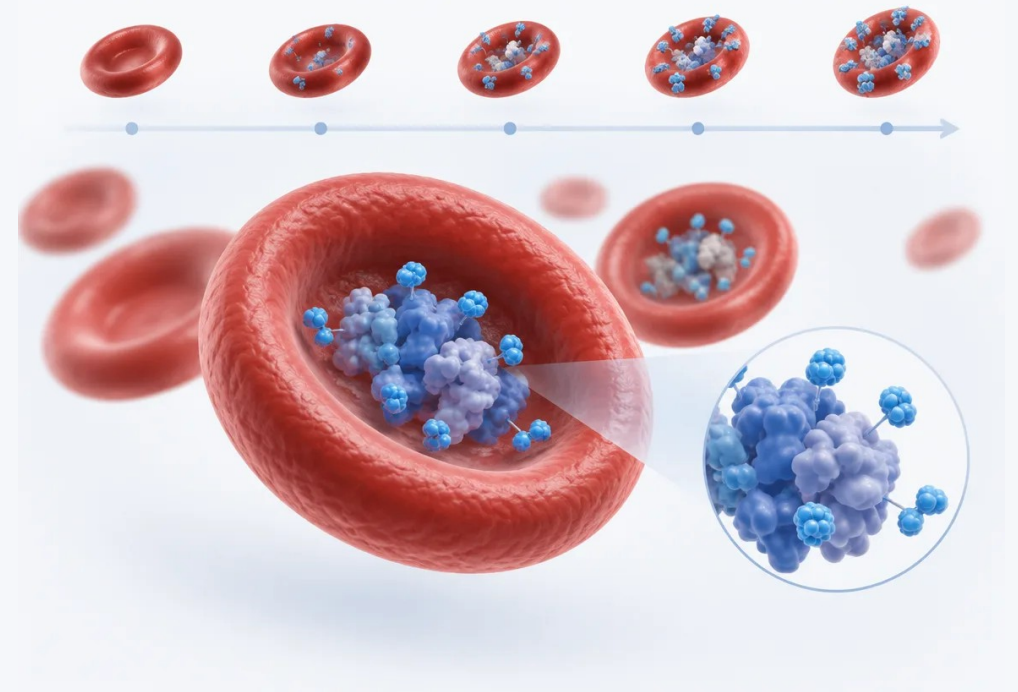


● 91 current ● 4 overdue ● 21 never tested

Each dot is one patient in the synthetic cohort, measured on Aug 23, 2026.

A1C turns glucose exposure over time into a measurable clinical signal

- 1 Glucose circulates in the bloodstream.
- 2 Some of it attaches to hemoglobin inside red blood cells.
- 3 A1C measures the percentage of hemoglobin with glucose attached.
- 4 Red blood cells live about 120 days, so A1C reflects roughly the previous 2–3 months.



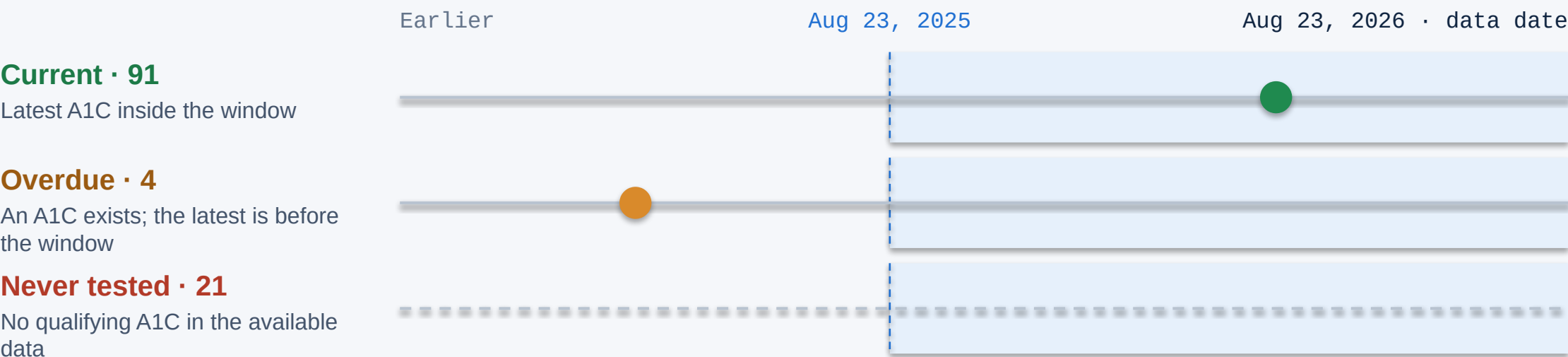
One A1C result is a point in time

Monitoring requires knowing when the latest result occurred.



From an A1C result to a care gap

Every patient is checked against the same 365-day window, ending on the fixed data date.



Exactly 365 days old = Current · 366 days = Overdue

Monitoring status is not a diagnosis and does not determine treatment.

The clinical question sounds simple. The data problem is not.

SIX SOURCE TABLES · ROWS

Patients	1,159
Conditions	40,811
Encounters	67,795
Observations	870,510
Medications	59,273
Procedures	187,126

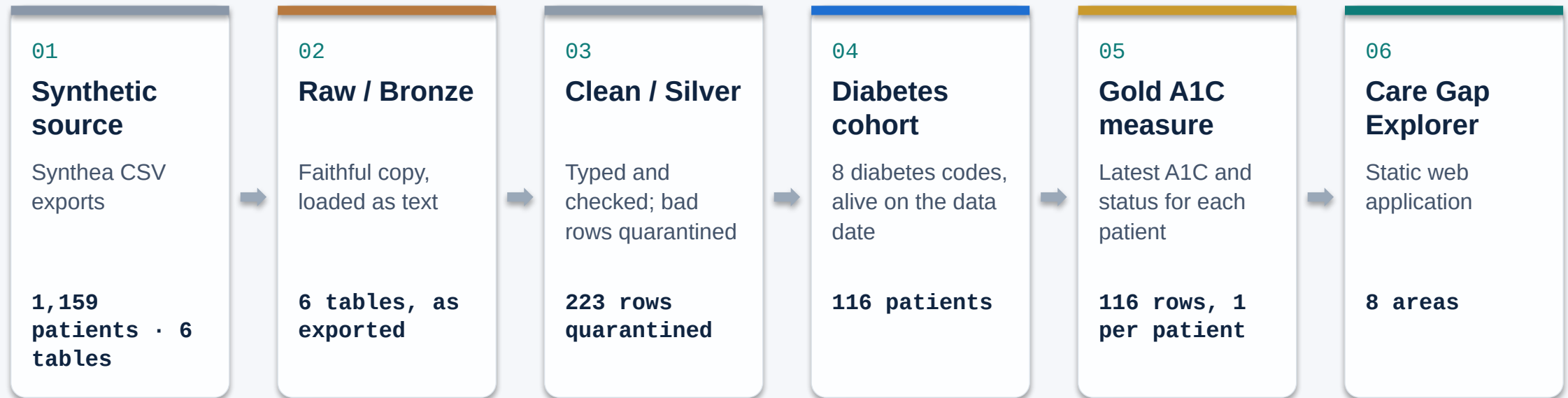
QUESTIONS THE PIPELINE MUST ANSWER

Who belongs in the diabetes cohort?	Which A1C belongs to which patient?
Which result is the latest?	What if no A1C exists?
How do we avoid duplicate patient rows?	What date should the measure use?

Reliable care-gap identification depends on reliable data engineering.

Six stages from synthetic source to application

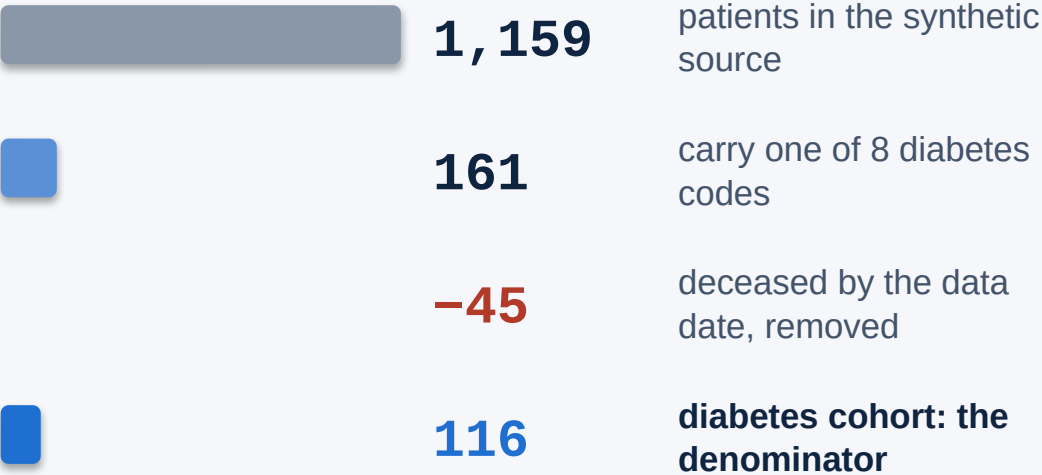
A medallion pipeline: each stage has one job and hands the next a smaller, cleaner set.



Every source table reconciles: **source rows = Silver rows + quarantined rows**

Building the denominator

FROM SOURCE TO COHORT



116 PATIENTS BY MONITORING STATUS



116 = 91 + 25

25 OPEN GAPS BY TYPE



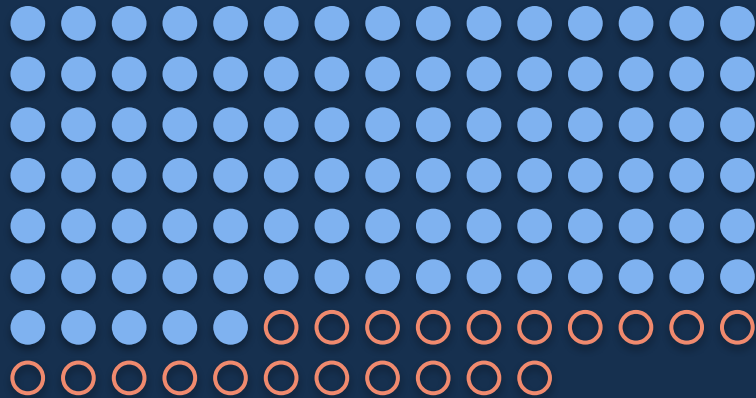
21 never tested · 4 overdue

25 = 21 + 4

Never tested means no qualifying A1C in the available data, not proof that a patient was never tested elsewhere.

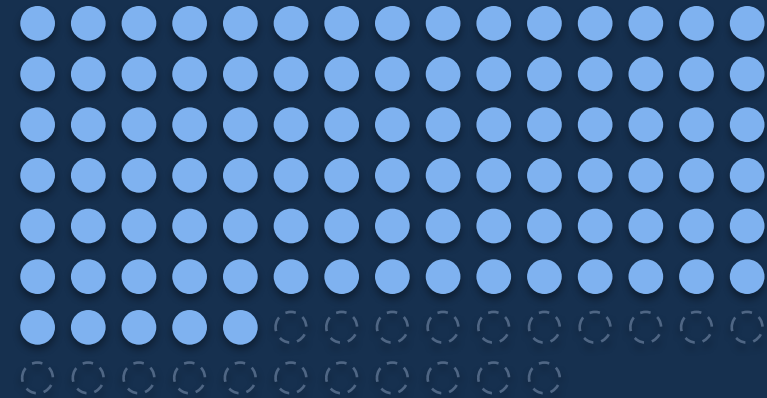
One join decision can hide most of the care gaps

`cohort LEFT JOIN latest_a1c`



116 rows · 21 kept with A1C = NULL

`cohort INNER JOIN latest_a1c`



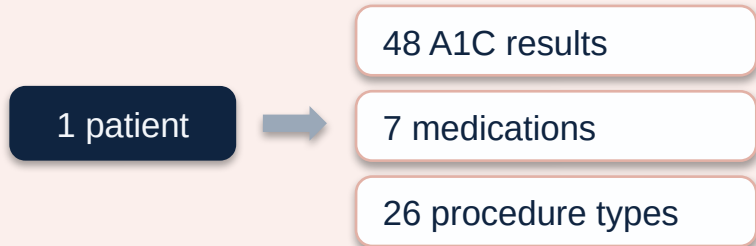
95 rows · 21 never-tested patients gone

Both queries run without an error. The inner join returns a plausible-looking table while deleting the patients we are trying to find.

NULL is part of the signal.

Reduce each event table to one row per patient first

JOINED AS EVENTS



$$48 \times 7 \times 26 = 8,736$$

rows for one synthetic patient (MRN 79bae4c9), if those tables are joined directly. Every count built on them inflates.

AGGREGATE · RANK · SELECT

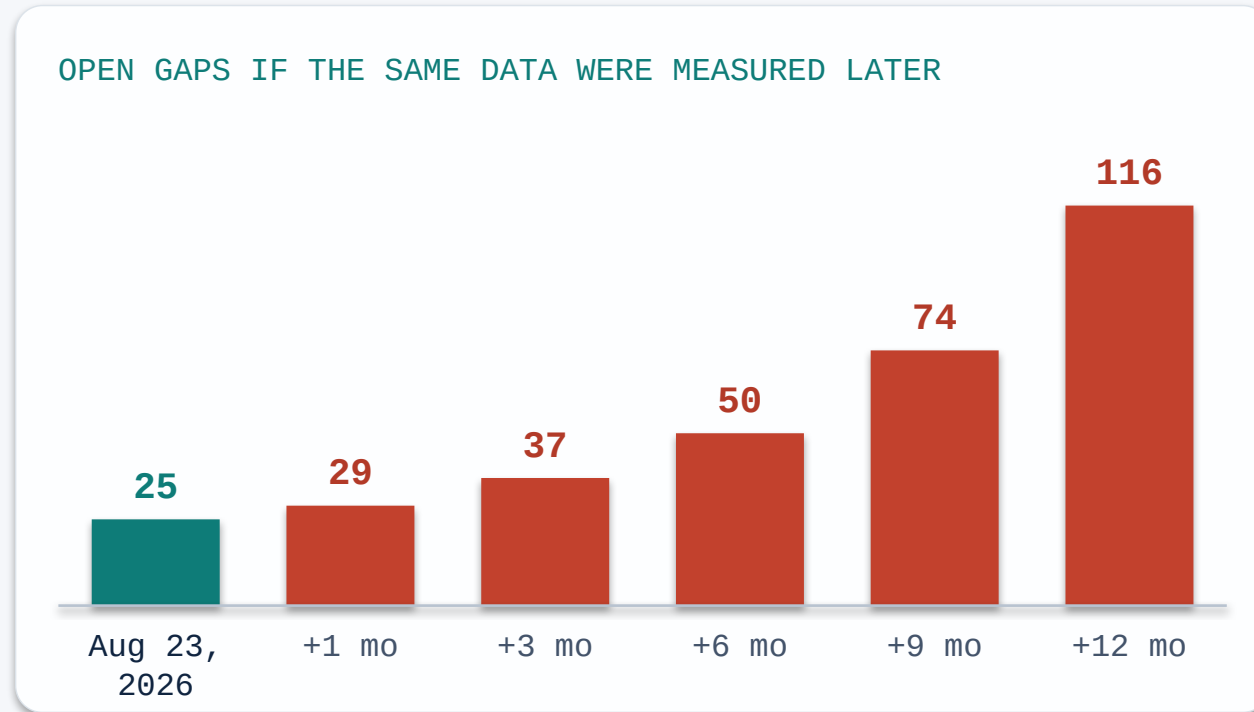
```
FROM a1c
QUALIFY row_number() OVER (
  PARTITION BY patient_id
  ORDER BY observed_at DESC
) = 1
```

The latest date and its value come from the same observation.
Audited: no patient has two results at the latest timestamp.

1 row per patient

Gold measure: 116 rows · 116 patients

Measure a frozen dataset against a fixed date



Fixed: Aug 23, 2026

The day the source data ends. The same data gives the same answer every time.



CURRENT_DATE

The calendar moves while the data stays frozen. Twelve months on, all 116 patients look overdue.

Reproducibility requires measuring the dataset against a known date.

Every number reconciles to patient-level data

$$116 = 91 + 25$$

cohort = current + open gaps

$$25 = 21 + 4$$

open gaps = never tested + overdue

Checked against the exported patient rows every time the application builds.

✓ **Patient identifiers present**

None missing

✓ **One row per patient**

116 rows, 116 patients

✓ **A1C values plausible**

All within 2.0–20.0%

✓ **No results after the data date**

None after Aug 23, 2026

✓ **Duplicate observations checked**

0 duplicate A1C results

✓ **Latest result deterministic**

0 ties at the latest timestamp

✓ **Cohort membership validated**

161 coded – 45 deceased = 116

✓ **Missing-A1C patients preserved**

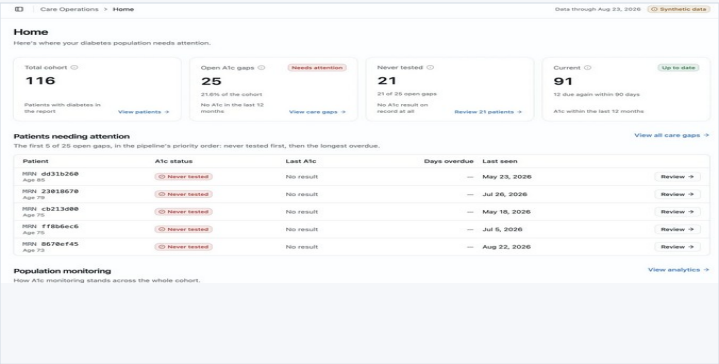
21 kept with no result

✓ **Source rows reconcile**

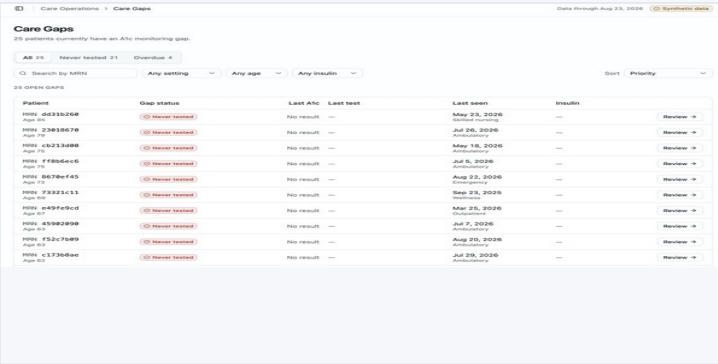
6 of 6 tables: source = Silver + quarantine

The Gold measure, put to work

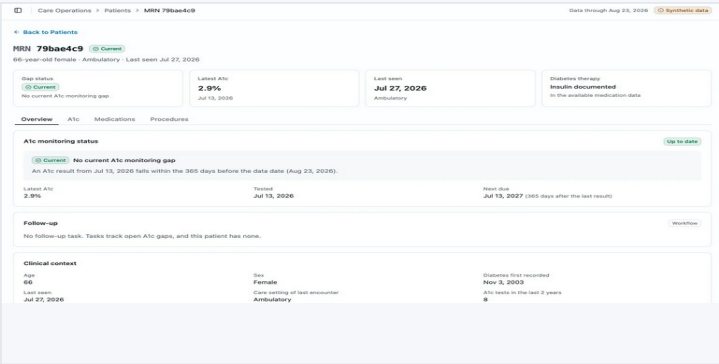
01 Identify Home



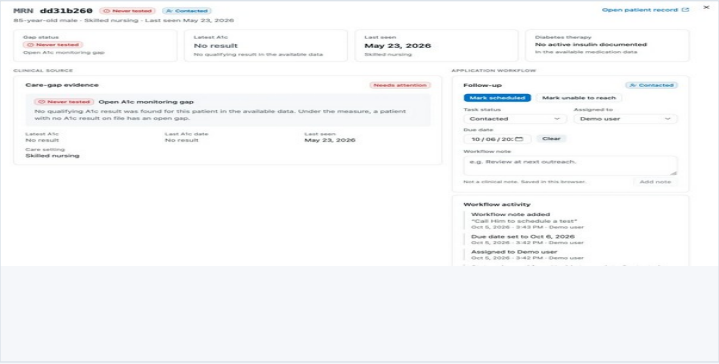
02 Prioritize Care Gaps



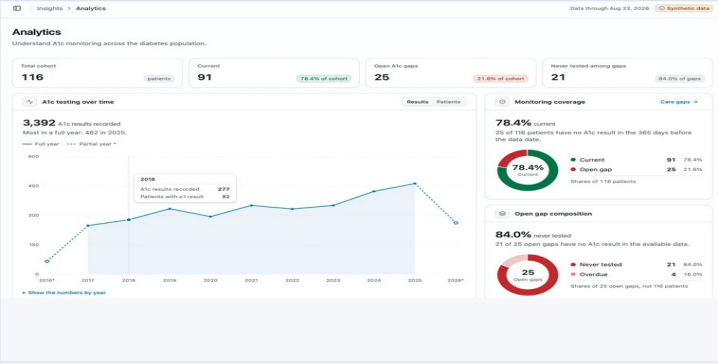
03 Investigate Patient workspace



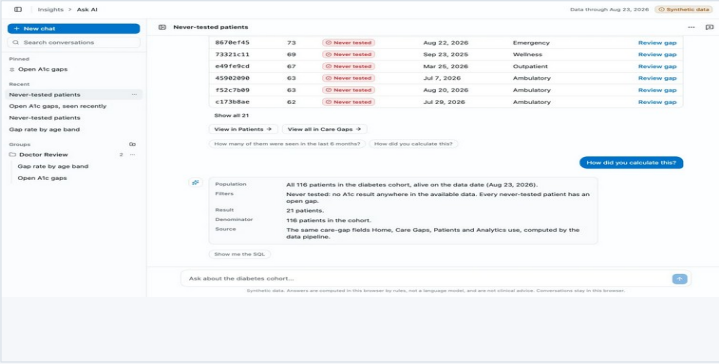
04 Track Tasks



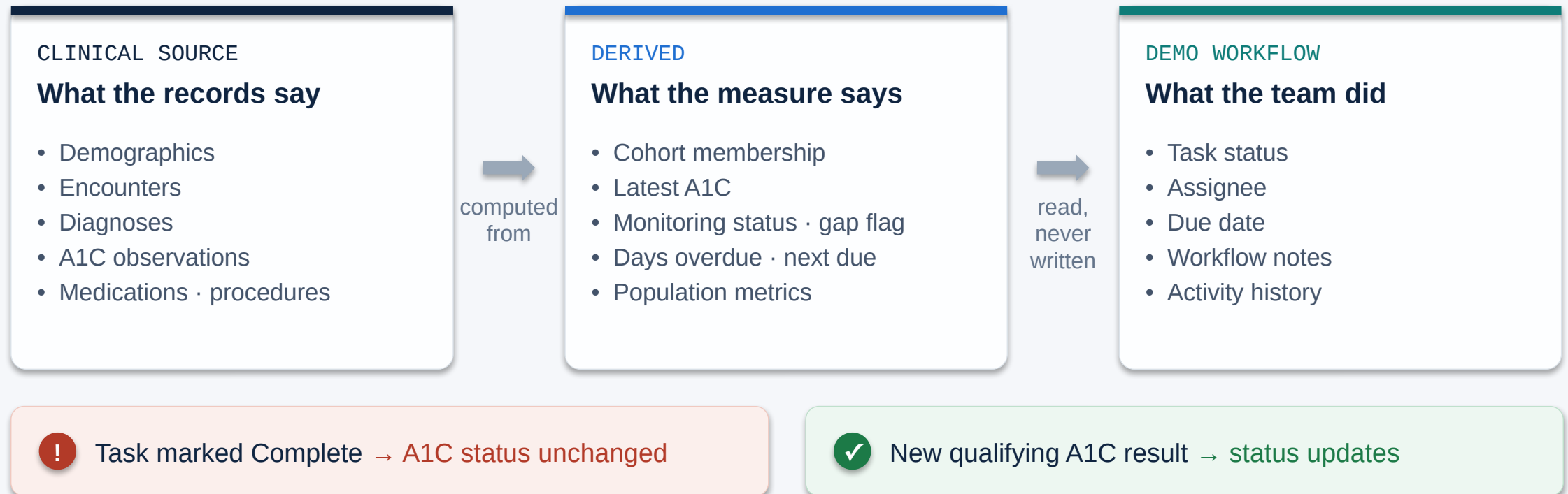
05 Analyze Analytics



06 Explain Ask AI



Operational workflow should not rewrite clinical truth



What I would change for production

PORTFOLIO, TODAY

- Synthetic CSV (Synthea)
- Local Python + DuckDB pipeline
- Patient-level Gold dataset
- Static web application



PRODUCTION

- | | |
|----------------------------------|--------------------------------|
| ✓ FHIR / EHR / warehouse sources | ✓ Orchestration |
| ✓ dbt / SQL transformations | ✓ Automated data-quality tests |
| ✓ PHI and security controls | ✓ Role-based access |
| ✓ Audit logging | ✓ Monitoring and alerting |
| ✓ Incremental processing | ✓ Semantic / measure layer |

The project demonstrates the full path from raw healthcare data to a reproducible, validated and explainable operational measure.